

AI QA Studio

Generated Test Cases Report

Generated On : 06-08-2026 09:42 PM

TC-001

Category : Positive

Priority : High

Scenario

Verify successful login with valid username and password

Preconditions

Precondition 1: User account exists in an active state.

Precondition 2: Browser is open and navigated to the login page URL.

Steps

Step 1: {'action': 'Navigate to the application login page.', 'expectedResult': 'Login page renders successfully showing Username field, Password field, and Login button.'}

Step 2: {'action': 'Enter valid registered username into the Username field.', 'expectedResult': 'Username field accepts and displays the entered string.'}

Step 3: {'action': 'Enter valid corresponding password into the Password field.', 'expectedResult': 'Password field accepts input and masks characters with dots or asterisks.'}

Step 4: {'action': 'Click the Login button.', 'expectedResult': 'User is successfully authenticated and redirected to the main dashboard.'}

Test Data

Username: valid_user@example.com, Password: ValidPassword123!

Expected Result

Design Technique

Equivalence Partitioning

TC-002

Category : Positive

Priority : High

Scenario

Verify login submission using Keyboard Enter key

Preconditions

Precondition 1: User account exists in an active state.

Precondition 2: Browser is open and navigated to the login page URL.

Steps

Step 1: {'action': 'Navigate to the application login page.', 'expectedResult': 'Login page is loaded.'}

Step 2: {'action': 'Enter valid username into the Username field.', 'expectedResult': 'Username text is correctly entered.'}

Step 3: {'action': 'Enter valid password into the Password field.', 'expectedResult': 'Password text is entered and masked.'}

Step 4: {'action': 'Press the Enter key on the keyboard while focused on Password field.', 'expectedResult': 'Login request is submitted, authenticating user and redirecting to the dashboard.'}

Test Data

Username: valid_user@example.com, Password: ValidPassword123!

Expected Result

Design Technique

Use Case Testing

TC-003

Category : Positive

Priority : Low

Scenario

Verify sequential focus navigation using TAB key

Preconditions

Precondition 1: Browser is open and navigated to the login page URL.

Steps

Step 1: {'action': 'Navigate to the login page.', 'expectedResult': 'Login page loads and initial focus is set to the body or Username field.'}

Step 2: {'action': 'Press the TAB key.', 'expectedResult': 'Focus moves directly to the Username input field.'}

Step 3: {'action': 'Press the TAB key again.', 'expectedResult': 'Focus shifts from Username field to the Password input field.'}

Step 4: {'action': 'Press the TAB key again.', 'expectedResult': 'Focus shifts from Password field to the Login button.'}

Test Data

No specific test data required.

Expected Result

Design Technique

State Transition

TC-004

Category : Negative

Priority : High

Scenario

Verify validation errors when submitting empty login form

Preconditions

Precondition 1: Browser is open and navigated to the login page URL.

Steps

Step 1: {'action': 'Navigate to the login page.', 'expectedResult': 'Login page displays with blank input fields.'}

Step 2: {'action': 'Click the Login button without entering credentials.', 'expectedResult': "Validation error messages 'Username is required' and 'Password is required' are displayed below respective fields."}

Test Data

Username: empty, Password: empty

Expected Result

Design Technique

Equivalence Partitioning

TC-005

Category : Negative

Priority : High

Scenario

Verify error message when entering invalid password for valid username

Preconditions

Precondition 1: User account exists in active state.

Precondition 2: Browser is open and navigated to the login page URL.

Steps

Step 1: {'action': 'Navigate to the login page.', 'expectedResult': 'Login page displays.'}

Step 2: {'action': 'Enter valid username into Username field.', 'expectedResult': 'Username field contains entered username.'}

Step 3: {'action': 'Enter incorrect password into Password field.', 'expectedResult': 'Password field displays masked characters.'}

Step 4: {'action': 'Click the Login button.', 'expectedResult': 'Error message 'Invalid credentials' is displayed and user remains on login page.'}

Test Data

Username: valid_user@example.com, Password: IncorrectPassword123

Expected Result

Design Technique

Equivalence Partitioning

TC-006

Category : Negative

Priority : High

Scenario

Verify error message when entering non-existent username

Preconditions

Precondition 1: Browser is open and navigated to the login page URL.

Steps

Step 1: {'action': 'Navigate to the login page.', 'expectedResult': 'Login page loads successfully.'}

Step 2: {'action': 'Enter a non-registered username into Username field.', 'expectedResult': 'Username field contains non-existent username.'}

Step 3: {'action': 'Enter any password into Password field.', 'expectedResult': 'Password field displays masked characters.'}

Step 4: {'action': 'Click the Login button.', 'expectedResult': 'Error message 'Invalid credentials' is displayed without revealing if username exists.'}

Test Data

Username: non_existent_user@example.com, Password: SomePassword123!

Expected Result

Design Technique

Boundary Value Analysis

TC-007

Category : Negative

Priority : High

Scenario

Verify account lockout after multiple consecutive failed attempts

Preconditions

Precondition 1: User account exists and active lockout threshold is 3 failed attempts.

Precondition 2: Browser is open and navigated to the login page URL.

Steps

Step 1: {'action': 'Navigate to the login page.', 'expectedResult': 'Login page is loaded.'}

Step 2: {'action': 'Enter valid username and invalid password, then click Login (Attempt 1).', 'expectedResult': 'Error message 'Invalid credentials' is displayed.'}

Step 3: {'action': 'Enter valid username and invalid password, then click Login (Attempt 2).', 'expectedResult': "Error message 'Invalid credentials' is displayed."}

Step 4: {'action': 'Enter valid username and invalid password, then click Login (Attempt 3).', 'expectedResult': "Lockout message 'Account locked due to multiple failed attempts' is displayed."}

Step 5: {'action': 'Enter valid username and correct password, then click Login.', 'expectedResult': "Access is blocked with message 'Account is locked. Please contact support'."}

Test Data

Username: valid_user@example.com, Password: WrongPassword!

Expected Result

Design Technique

State Transition

TC-008

Category : Negative

Priority : High

Scenario

Verify protection against SQL Injection in input fields

Preconditions

Precondition 1: Browser is open and navigated to the login page URL.

Steps

Step 1: {'action': 'Navigate to the login page.', 'expectedResult': 'Login page renders correctly.'}

Step 2: {'action': "Enter SQL Injection script `` OR '1'='1` in Username field.", 'expectedResult': 'Text is entered into field without execution.'}

Step 3: {'action': "Enter SQL Injection script `` OR '1'='1` in Password field.", 'expectedResult': 'Text is entered into field masked.'}

Step 4: {'action': 'Click the Login button.', 'expectedResult': "System rejects input with generic error message 'Invalid credentials' without database execution."}

Test Data

Username: ' OR '1'='1, Password: ' OR '1'='1

Expected Result

Design Technique

Error Guessing

TC-009

Category : Negative

Priority : Medium

Scenario

Verify system handling of Cross-Site Scripting (XSS) payload in username field

Preconditions

Precondition 1: Browser is open and navigated to the login page URL.

Steps

Step 1: {'action': 'Navigate to the login page.', 'expectedResult': 'Login page renders properly.'}

Step 2: {'action': "Enter XSS string `alert('XSS')` into Username field.", 'expectedResult': 'String is accepted as plain text.'}

Step 3: {'action': 'Enter valid password and click Login button.', 'expectedResult': 'Login fails safely, error message displays sanitized output, and no alert script executes.'}

Test Data

Username: alert('XSS'), Password: ValidPassword123!

Expected Result

Design Technique

Error Guessing